



CANDIDATE

DAT246-0044-LBO_2025-10-
27_Bonus_NA

TEST

DAT246 HT25 LP1 Ordinarie
tentamen - J

Subject code	--
Evaluation type	1
Test opening time	27.10.2025 14:00
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Exam information

Question	Question title	Status	Marks	Question type
i	DAT246-251027 Exam Information			Information or resources

Empirical Software Engineering

Question	Question title	Status	Marks	Question type
1	ABC	Answered	3/3	Text area
2	Obtrusiveness (Matching)	Partially Correct	1/4	Matching
3	Case Study Time Frame	Answered	1/4	Text area
4	Triangulation	Answered	0/3	Text area
5	Experimentation Terminology (Matching)	Correct	5/5	Matching
6	Sampling	Answered	0/4	Text area
7	Threats to Validity	Answered	2/6	Text area

Bayesian Data Analysis

Question	Question title	Status	Marks	Question type
8	Compulsory Steps of BDA	Answered	4/8	Text area
9	Types of Relationships	Answered	8/9	Text area
10	Transformations in Regressions	Answered	1/5	Text area
11	Outcome Distribution: Count (MC)	Partially Correct	0/4	Multiple Response
12	Adjustment Set Identification	Partially Correct	0/3	Multiple Response
13	Model Derivation	Answered	4/12	Text area

1 ABC

What do A, B, and C mean in the “ABC of Software Engineering Research” by Klaas-Jan Stol and Brian Fitzgerald [1]?

Write your answer in the box below. Changes are saved automatically.

A - Actors -> Generalizability
B - Behaviors -> Precision
C - Context -> Realism

[1] Stol, K. J., & Fitzgerald, B. (2018). The ABC of software engineering research. *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 27(3), 1-51.

Maximum marks: 3

2 Obtrusiveness (Matching)

Klaas-Jan Stol and Brian Fitzgerald structure eight research strategies along two axes. One axis represents the “level of obtrusiveness”. Rank the following research strategies regarding their typical level of obtrusiveness as presented by Klaas-Jan Stol and Brian Fitzgerald [1].

Please match the values:

	Most obtrusive	Very obtrusive	Slightly obtrusive	Least obtrusive
Computer Simulation	☐	☐	☐	☒
Field Experiment	☐	☒	☒	☐
Sample Study	☒	☐	☒	☐
Laboratory Experiment	☒	☒	☐	☐

[1] Stol, K. J., & Fitzgerald, B. (2018). The ABC of software engineering research. *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 27(3), 1-51.

Maximum marks: 4

3 Case Study Time Frame

Explain the difference between a **retrospective** and **longitudinal** case study. Name one unique advantage and disadvantage for each of them.

Write your answer in the box below. Changes are saved automatically.

They are different from time frame.
Retrospective case study provides interval time frame, which is good for periodic case study observation.
Logitudinal case study provides continuous time frame, which is good for long term case study observation.

Maximum marks: 4

4 Triangulation

Explain what triangulation in case study research is and what it is used for. Describe one type of triangulation.

Write your answer in the box below. Changes are saved automatically.

Triangulation is a case study method.
It is used for case study research to stablize and iterate.
One type of triangulation is objective process.

Maximum marks: 3

5 Experimentation Terminology (Matching)

In a (fictional) study, researchers investigated whether the use of ChatGPT helps university students solve coding assignments faster. Match the following terms relevant to experimentation to their respective entity in the study: treatment, control, outcome, subject, object.

Please match the values:

	outcome	subject	treatment	control	object
not using ChatGPT	☐	☐	☐	☒	☐
coding assignment	☐	☐	☐	☐	☒
duration of the assignment	☒	☐	☐	☐	☐
university student	☐	☒	☐	☐	☐
using ChatGPT	☐	☐	☒	☐	☐

Maximum marks: 5

6 Sampling

Explain the difference between probabilistic and non-probabilistic sampling. Give one example of a probabilistic and one of a non-probabilistic sampling method.

Write your answer in the box below. Changes are saved automatically.

Probabilistic sampling: Sample Studies
Non-probabilistic sampling: Field Studies

Maximum marks: 4

7 Threats to Validity

Name the four types of threats to validity that are typically discussed in software engineering research. Explain, how each of these types threatens the validity of research results.

Write your answer in the box below. Changes are saved automatically.

Internal Validity, External Validity, Context Validity, Realism Validity

Internal Validity shows what the causal reasoning is in generalizability.
External Validity shows the realism of context

Experimental experiments have high internal validity and low external validity.
Observational experiments have low internal validity and high external validity.

Maximum marks: 6

8 Compulsory Steps of BDA

Name and describe compulsory steps in a full Bayesian data analysis in the right order.

Write your answer in the box below. Changes are saved automatically.

Find the causal structure -> DAG -> Prior -> Prior detective checks -> Fit the model ->
Diagnostics (MCMC) -> Posterior Detective Checks -> Inference (iterative)

Maximum marks: 8

9 Types of Relationships

In a directed acyclic graph (DAG), there are three major ways how a variable z could be related to two variables $x \rightarrow y$. For each of the three ways,

- specify how the variable would be related to x and y (using $\rightarrow$ as an arrow symbol),
- state how this relationship is commonly called, and
- explain how conditioning on z would affect the estimation of the effect of x on y .

Write your answer in the box below. Changes are saved automatically.

Fork (Confounder): $x \leftarrow z \rightarrow y$

Condition on z gives the total effect of x on y .

Condition on z gives the direct effect of x on y .

Pipe (Mediator): $x \rightarrow z \rightarrow y$

Do not condition on z gives the total effect of x on y .

Condition on z gives the direct effect of x on y .

Collider: $x \rightarrow z \leftarrow y$

Condition on z gives no effect of x on y .

Decendant: z inherits from a variable

Condition on z copies the effect behavior of the variable it inherits from.

Maximum marks: 9

10 Transformations in Regressions

Some regression models have the following form.

$$y \sim \text{Binomial}(n, p)$$
$$f(p) \leftarrow \alpha + \beta_x x + \beta_z z$$

Explain, what $f()$ generally does, and how it is usually called. Then, name the respective function that would be appropriate in the given example above and explain, how this particular function works.

Write your answer in the box below. Changes are saved automatically.

Binomial distribution gives the probability of p in n trials.

p - probability

n - number of trials

Link function f(p) gives the probability

alpha - the intercept

beta - slope, gradient of the variables

It shows that predictors x and z have effect on outcome y.

Maximum marks: 5

11 Outcome Distribution: Count (MC)

Which of the following distribution families have maximum entropy for count variables?

Select one or more alternatives:

☐ Negative-Binomial Distribution



☐ Binomial Distribution



☐ Log-Normal Distribution

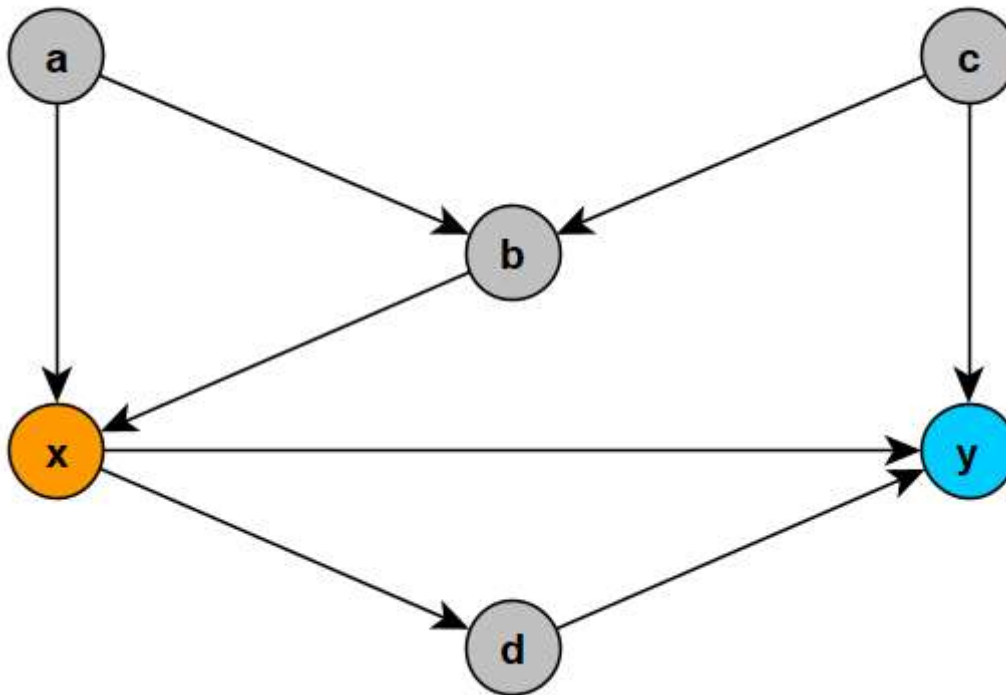
☒ Poisson Distribution



Maximum marks: 4

12 Adjustment Set Identification

Given the following DAG, select all valid adjustment sets to estimate the unbiased (total or direct) effect of x on y.



Select one or more alternatives:

☐ { a; b; d }



☐ { b ; d }

☐ { a; c }

☒ { c }



Maximum marks: 3

13 Model Derivation

Consider a university course with multiple assignments. You want to investigate if **passing an assignment** is affected by **the time spent on it**. You have a data set from with one row per combination of student and assignment containing the following variables:

Variable	Description	Type
student	Unique identifier of the student	ID
assignment	Unique identifier of the assignment	ID
time.spent	Time spent on the respective assignment	Standardized $N(0,1)$
assignment.passed	Whether or not the student passed the assignment	Boolean
interest	Level of interest of the student in the topic of the respective assignment	Standardized $N(0,1)$

Through identifying the adjustment set from a DAG, you have already determined that the variable **interest** confounds the effect of interest.

Specify two statistical models that estimates the unbiased effect

time.spent → **assignment.passed** where the outcome distribution has maximum entropy. The models must consist of

1. the outcome variable modeled with a probability distribution that has maximum entropy,
2. a linear model that describes the shape-defining parameter of that distribution where
 1. the linear model contains all predictors necessary to estimate an unbiased effect of interest, and
 2. the shape-defining parameter only accepts valid values, and finally
3. feasible prior distributions for all coefficients in the model.

In the first model, use *complete pooling*. In the second model, use *partial pooling* on **interest** based on the fact that some students are generally more interested than others. Finally, explain the potential benefit of partial pooling.

Write your answer in the box below. Changes are saved automatically.

Model 1:
time.spent -> assignment. passed

assignment.passed ~ Normal (mu, sigma)

linear model:
mu = alpha + beta_time.spend * time.spent_i

Model 2:
interest -> time.spent
interest -> assignment.passed
time.spent -> assignment. passed

assignment.passed ~ Normal (mu, sigma)

linear model:
mu = alpha + beta_time.spend * time.spent_i + beta_interest * interest_i

time.spend has normal distribution,
interest has normal distribution,
assignment.passed has binomial distribution.

You can either write the model in mathematical notation (using characters like $_$, $\sim$ and $\leftarrow$) or as you would specify the *alist* object in the *rethinking* package.

Maximum marks: 12